

Course: EEE F311 Communication Systems
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1 Objectives

- Figure of Merit, BER

2 Tutorial Problems

1. In an analog communication system, the demodulation gain is defined as the ratio of the SNR at the output of the demodulator to the SNR at the output of the noise-limiting filter at the receiver front end. Find expression for the demodulation gain of SSB.
2. Find the threshold for OOK (On-Off Keying) detection with unequal probability of transmission (p_1 for bit '1' and p_0 for bit '0') under AWGN with mean $\mu = 0$ and variance σ^2 . Take maximum signal amplitude for the pulse as A_p .

Solution

1. Proven in lecture: Bandpass noise model $n(t) = n_i \cos 2\pi f_c t - n_q \sin 2\pi f_c t$

$$y(t) = Am(t) \cos 2\pi f_c t \pm Am_h(t) \sin 2\pi f_c t + n_i \cos 2\pi f_c t - n_q \sin 2\pi f_c t$$

$$P_{si} = A^2 P_m$$

$$\text{Proven in lecture: } P_{ni} = P_{ni} = P_{nq} = P_n$$

$$\text{SNR at the input } \gamma_i = \frac{A^2 P_m}{P_n}$$

Synchronous detector (Multiply by $\cos 2\pi f_c t$ and then LPF)

$$r(t) = \frac{A}{2} m(t) + \frac{1}{2} n_i(t)$$

$$P_{s0} = \frac{A^2}{4} P_m$$

$$P_{n0} = \frac{1}{4} P_{ni} = \frac{1}{4} P_n$$

$$\text{SNR at the output } \gamma_0 = \frac{A^2 P_m}{P_n}$$

$$\text{Figure of Merit: } \frac{\gamma_0}{\gamma_i} = 1$$

2. Assume a threshold value λ .

$$\text{BER for bit '1': } P_{e1} = P(A_p + n < \lambda) = Q\left(\frac{A_p - \lambda}{\sigma}\right)$$

$$\text{BER for bit '0': } P_{e0} = P(n > \lambda) = Q\left(\frac{\lambda}{\sigma}\right)$$

$$\text{Average BER: } P_e = p_1 P_{e1} + p_0 P_{e0}$$

To get λ :

Differentiate P_e wrt to λ (Use Leibnitz rule) and to get the minima, we have

$$\frac{dP_e}{d\lambda} = p_1 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(A_p - \lambda)^2}{2\sigma^2}} - p_0 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\lambda)^2}{2\sigma^2}} = 0$$

$$\text{Solving above } \lambda = \frac{A_p}{2} + \frac{\sigma^2}{A_p} \log_e\left(\frac{p_0}{p_1}\right)$$